Menu:

```
===== MULTI-UTILITY TOOLKIT =====
1. Datetime and Time Operations
2. Mathematical Operations
3. Random Data Generation
4. Generate Unique Identifiers (UUID)
5. File Operations (Custom Module)
6. Explore Module Attributes (dir())
7. Exit
Enter your choice: 1
```

Datetime and Time Operations:

```
--- DATE & TIME MENU ---
1. Current Date & Time
2. Custom Format
3. Difference Between Dates
4. Stopwatch
5. Countdown Timer
6. Back
Enter choice: 2
05-02-2026 14:58:15
========================

--- DATE & TIME MENU ---
1. Current Date & Time
2. Custom Format
3. Difference Between Dates
4. Stopwatch
5. Countdown Timer
6. Back
Enter choice: 3
Enter first date (DD-MM-YYYY): 22-09-2026
Enter second date (DD-MM-YYYY): 10-10-2026
Difference: 18 days
========================

--- DATE & TIME MENU ---
1. Current Date & Time
2. Custom Format
3. Difference Between Dates
4. Stopwatch
5. Countdown Timer
6. Back
Enter choice: 4
Press Enter to start stopwatch...
Press Enter to stop...
Elapsed Time: 5.57 seconds
========================
```

```
--- DATE & TIME MENU ---
1. Current Date & Time
2. Custom Format
3. Difference Between Dates
4. Stopwatch
5. Countdown Timer
6. Back
Enter choice: 5
Enter seconds: 10
Time left: 1 secc
Time's up!
========================

--- DATE & TIME MENU ---
1. Current Date & Time
2. Custom Format
3. Difference Between Dates
4. Stopwatch
5. Countdown Timer
6. Back
Enter choice: 6
```

Mathematical Operations:

```
--- MATH MENU ---
1. Calculate Factorial
2. Solve Compound Interest
3. Trigonometric Calculations
4. Area of Geometric shapes
5. Back
Enter choice: 1
Enter number: 4
Factorial: 24
==========================

--- MATH MENU ---
1. Calculate Factorial
2. Solve Compound Interest
3. Trigonometric Calculations
4. Area of Geometric shapes
5. Back
Enter choice: 2
Principal: 2400
Rate (%): 2
Time (years): 1.5
Amount: 2472.36
==========================

--- MATH MENU ---
1. Calculate Factorial
2. Solve Compound Interest
3. Trigonometric Calculations
4. Area of Geometric shapes
5. Back
Enter choice: 3
Enter angle (in radians): 60
{'sin': -0.3, 'cos': -0.95, 'tan': 0.32}
==========================
```

```
--- MATH MENU ---
1. Calculate Factorial
2. Solve Compound Interest
3. Trigonometric Calculations
4. Area of Geometric shapes
5. Back
Enter choice: 4

1. Circle
2. Rectangle
Choose: 1
Radius: 15
Area of circle:  706.86
========================

--- MATH MENU ---
1. Calculate Factorial
2. Solve Compound Interest
3. Trigonometric Calculations
4. Area of Geometric shapes
5. Back
Enter choice: 4

1. Circle
2. Rectangle
Choose: 2
Length: 15
Breadth: 5
Area of rectangle:  75.0
========================
```

Random Data Generation:

```
--- RANDOM MENU ---
1. Generate Random Number
2. Generate Random List
3. Create Password
4. Generate OTP
5. Random Sampling from Dataset
6. Game (Dice)
7. Back
Enter choice:1
Random Number:  35
========================
```

```
--- RANDOM MENU ---
1. Generate Random Number
2. Generate Random List
3. Create Password
4. Generate OTP
5. Random Sampling from Dataset
6. Game (Dice)
7. Back
Enter choice:2
Random List:  [24, 1, 16, 44, 47]
========================

--- RANDOM MENU ---
1. Generate Random Number
2. Generate Random List
3. Create Password
4. Generate OTP
5. Random Sampling from Dataset
6. Game (Dice)
7. Back
Enter choice:3
Password:  HfovUavy
========================

--- RANDOM MENU ---
1. Generate Random Number
2. Generate Random List
3. Create Password
4. Generate OTP
5. Random Sampling from Dataset
6. Game (Dice)
7. Back
Enter choice:4
OTP:  411581
========================
```

```
--- RANDOM MENU ---
1. Generate Random Number
2. Generate Random List
3. Create Password
4. Generate OTP
5. Random Sampling from Dataset
6. Game (Dice)
7. Back
Enter choice:5
Enter Data: 23435678
Sample:  ['7', '6', '8']
========================

--- RANDOM MENU ---
1. Generate Random Number
2. Generate Random List
3. Create Password
4. Generate OTP
5. Random Sampling from Dataset
6. Game (Dice)
7. Back
Enter choice:6
Dice Number:  4
========================
```

Generate Unique Identifiers (UUID):

```
===== MULTI-UTILITY TOOLKIT =====
1. Datetime and Time Operations
2. Mathematical Operations
3. Random Data Generation
4. Generate Unique Identifiers (UUID)
5. File Operations (Custom Module)
6. Explore Module Attributes (dir())
7. Exit
Enter your choice: 4

--- GENERATE UNIQUE IDENTIFIERS ---
Generated UUID: 27d3e0e1-3022-407a-85c7-1975c5073e94
=========================
```

File Operations (Custom Module)

```
--- FILE MENU ---
1. Create File
2. Write File
3. Read File
4. Append File
5. Back
Enter choice: 1
Enter file name: journal.txt
File created successfully

--- FILE MENU ---
1. Create File
2. Write File
3. Read File
4. Append File
5. Back
Enter choice: 2
Enter file name: journal.txt
Enter data: Hello. Today, I had a great day.
File written successfully

--- FILE MENU ---
1. Create File
2. Write File
3. Read File
4. Append File
5. Back
Enter choice: 3
Enter file name: journal.txt
Content: Hello. Today, I had a great day.
```

Explore Module Attributes (dir()):

```
===== MULTI-UTILITY TOOLKIT =====
1. Datetime and Time Operations
2. Mathematical Operations
3. Random Data Generation
4. Generate Unique Identifiers (UUID)
5. File Operations (Custom Module)
6. Explore Module Attributes (dir())
7. Exit
Enter your choice: 6
Enter module name: math_tools
['__builtins__', '__cached__', '__doc__', '__file__', '__loader__', '__name__', '__package__', '__spec__', 'area_circle', 'area_rectangle', 'compound_interest', 'factorial', 'logarithm'
, 'math', 'trigonometry']
=====================
```

Exit:

```
===== MULTI-UTILITY TOOLKIT =====
1. Datetime and Time Operations
2. Mathematical Operations
3. Random Data Generation
4. Generate Unique Identifiers (UUID)
5. File Operations (Custom Module)
6. Explore Module Attributes (dir())
7. Exit
Enter your choice: 7
-------------------------
Thank you for using Multi-Utility Toolkit
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```